



M135 Homework 2 Sol'n

$$\frac{f(t)}{t} = \int_0^{\infty} F(\omega) d\omega$$

$$\begin{aligned} 1) a) \mathcal{L}\left[\frac{\sin x}{x}\right] &= \int_p^{\infty} \frac{1}{t^2+1} dt \\ &= \lim_{R \rightarrow \infty} (\arctan t \Big|_p^R) \\ &= \lim_{R \rightarrow \infty} (\arctan R - \arctan p) \\ &= \pi/2 - \arctan(p) \end{aligned}$$

$$\begin{aligned} b) \mathcal{L}[t^n e^{at}] &= (-1)^n \frac{d^n}{dp^n} \left(\frac{1}{p-a} \right) \\ &= \frac{(-1)^{n+1} (n+1)}{(p-a)^{n+1}} \end{aligned}$$

$$2) a) \mathcal{L}^{-1} \left[\frac{1}{p^2(p^2+k^2)} \right]$$

$$\begin{aligned} \text{via PFD } \mathcal{L}^{-1} \left[\frac{1}{p^2(p^2+k^2)} \right] &= \mathcal{L}^{-1} \left[\frac{1}{k^2 p^2} - \frac{k}{k^2(p^2+k^2)} \right] \\ &= \frac{x}{k^2} - \frac{1}{k^3} \sin(kx) \text{ or } \frac{t}{k^2} - \frac{1}{k^3} \sin(kt) \end{aligned}$$

$$\begin{aligned} \text{via convolution } \mathcal{L}^{-1} \left[\frac{1}{k} \left(\frac{1}{p^2} \right) \left(\frac{k}{p^2+k^2} \right) \right] &= \frac{1}{k} x * \sin(kx) \\ &= \frac{1}{k} \int_0^x (x-\tau) \sin(k\tau) d\tau \quad \begin{array}{l} u=x-\tau \quad du=-d\tau \\ dv=\sin(k\tau) \quad v=-\frac{1}{k} \cos(k\tau) \end{array} \\ &= \frac{1}{k} \left(-\frac{1}{k} (x-\tau) \cos(k\tau) \Big|_0^x - \frac{1}{k} \int_0^x \cos(k\tau) d\tau \right) \\ &= \frac{1}{k^2} x + \left(-\frac{1}{k^3} \sin(k\tau) \Big|_0^x \right) \\ &= \frac{1}{k^2} x - \frac{1}{k^3} \sin(kx) \end{aligned}$$

$$\begin{aligned} b) \mathcal{L}^{-1} \left[\frac{1-2p}{p^2+4p+5} \right] &= \mathcal{L}^{-1} \left[\frac{1-2p}{(p+2)^2+1} \right] = \mathcal{L}^{-1} \left[\frac{-2(p+2)+5}{(p+2)^2+1} \right] \\ &= \mathcal{L}^{-1} \left[\frac{-2(p+2)}{(p+2)^2+1} \right] + 5 \mathcal{L}^{-1} \left[\frac{1}{(p+2)^2+1} \right] \end{aligned}$$

$$= -2e^{-2x} \cos(x) + 5e^{-2x} \sin x$$

$$c) \mathcal{L}^{-1} \left[\frac{8p^2 - 4p + 12}{p(p^2 + 4)} \right] = \mathcal{L}^{-1} \left[\frac{3}{p} + \frac{5p}{p^2 + 4} - \frac{2(2)}{p^2 + 4} \right]$$

$$= 3 + 5\cos(2x) - 2\sin(2x)$$

$$3) a) \begin{cases} y'' - 2y' + 2y = \cos t \\ y(0) = 1 \quad y'(0) = 0 \end{cases}$$

Let $y = \mathcal{L}[y(t)]$. Apply L.T.

$$p^2 y - py(0) - y'(0) - 2py + 2y(0) + 2y = \frac{1}{p^2 + 1}$$

$$(p^2 - 2p + 2)y - p - 2 = \frac{1}{p^2 + 1}$$

$$y = \frac{1}{(p^2 + 1)(p^2 - 2p + 2)} + \frac{p + 2}{p^2 - 2p + 2}$$

$$= \frac{1}{5} \left(\frac{2p + 1}{p^2 + 1} + \frac{-2p + 3}{(p - 1)^2 + 1} \right) + \frac{(p - 1) + 3}{(p - 1)^2 + 1}$$

$$= \frac{1}{5} \left(\frac{2p}{p^2 + 1} + \frac{1}{p^2 + 1} - \frac{2(p - 1)}{(p - 1)^2 + 1} + \frac{1}{(p - 1)^2 + 1} \right) + \frac{p - 1}{(p - 1)^2 + 1} + \frac{3}{(p - 1)^2 + 1}$$

$$= \frac{2}{5} \frac{p}{p^2 + 1} + \frac{1}{5(p^2 + 1)} + \frac{3}{5} \left(\frac{p - 1}{(p - 1)^2 + 1} \right) + \frac{16}{5} \left(\frac{1}{(p - 1)^2 + 1} \right)$$

Apply $\mathcal{L}^{-1}$

$$\rightarrow y(t) = \frac{2}{5} \cos t + \frac{1}{5} \sin t + \frac{3}{5} e^t \cos t + \frac{16}{5} e^t \sin t$$

$$b) \begin{cases} y'' + 2y' + y = 4e^{-t} \\ y(0) = 2 \quad y'(0) = -1 \end{cases}$$

Let $y = \mathcal{L}[y(t)]$. Apply $\mathcal{L}$

$$p^2 y - py(0) - y'(0) + 2py + 2y(0) + y = 4 \left(\frac{1}{p + 1} \right)$$

$$(p^2 + 2p + 1)y - 2p + 5 = 4 \left(\frac{1}{p + 1} \right)$$

$$y = \frac{4}{(p + 1)(p + 1)^2} + \frac{2p - 5}{(p + 1)^2} = \frac{4}{(p + 1)^3} + \frac{2(p + 1)}{(p + 1)^2} - \frac{7}{(p + 1)^2}$$

Apply $\mathcal{L}^{-1}$

$$y(t) = 2t^2 e^{-t} + 2e^{-t} - 7te^{-t}$$

$$4) \begin{cases} y^{(4)} - 4y''' + 6y'' - 4y' + y = 0 \\ y(0) = 0 \quad y'(0) = 1 \quad y''(0) = 0 \quad y'''(0) = 1 \end{cases}$$

$$\cancel{p^4 y - p^3 y(0) - p^2 y'(0) - p y''(0) - y'''(0)} - \cancel{4p^3 y + 4p^2 y(0) + 4p y'(0) + 4y''(0)} + \cancel{6p^2 y - 6p y(0) - 6y'(0)} - \cancel{4p y + 4y(0)} + y = 0$$

$$(p^4 - 4p^3 + 6p^2 - 4p + 1)y - p^2 - 1 + 4p - 6 = 0$$

$$y = \frac{p^2 - 4p + 7}{p^4 - 4p^3 + 6p^2 - 4p + 1} = \frac{p^2 - 4p + 7}{(p-1)^4} = \frac{1}{(p-1)^2} - \frac{2}{(p-1)^3} + \frac{4}{6(p-1)^4}$$

$$y(t) = te^t - t^2 e^t + \frac{4}{6} t^3 e^t$$

$$5) \mathcal{L}^{-1} \left[\frac{p}{(p^2+1)^2} \right]$$

a) Convolution

$$\mathcal{L}^{-1} \left[\left(\frac{p}{p^2+1} \right) \left(\frac{1}{p^2+1} \right) \right] = \cos x * \sin x = \int_0^x \cos(x-\tau) \sin \tau d\tau$$

$$= \frac{1}{2} \int_0^x [\sin(\tau + (x-\tau)) + \sin(\tau - (x-\tau))] d\tau$$

$$= \frac{1}{2} \int_0^x \sin x + \sin(2\tau - x) d\tau$$

$$= \frac{1}{2} \tau \sin x \Big|_0^x - \left[\frac{1}{4} \cos(2\tau - x) \right]_0^x$$

$$= \frac{1}{2} x \sin x$$

b) No Convolution you already have the answer :)

Recognize that $\frac{p}{(p^2+1)^2} = \frac{1}{2} \frac{d}{dp} \left(\frac{1}{p^2+1} \right)$

$$6) \begin{cases} y'' - 2y' + y = f(t) \\ y(0) = 0 \quad y'(0) = 1 \end{cases} \quad \text{Where } f(t) = \begin{cases} t & 1 \leq t < 2 \\ 0 & \text{else} \end{cases}$$

Apply L.T.

$$\begin{aligned} p^2 y - p y(0) - y'(0) - 2p y + 2y(0) + y &= \int_0^\infty e^{-pt} f(t) dt \\ (p^2 - 2p + 1)y - 1 &= \int_1^2 t e^{-pt} dt \end{aligned}$$

$u = t \quad du = e^{-pt}$
 $du = dt \quad v = -\frac{1}{p} e^{-pt}$

$$\begin{aligned} &= -\frac{t}{p} e^{-pt} \Big|_1^2 - \frac{1}{p} \int_1^2 e^{-pt} dt \\ &= -\frac{2}{p} e^{-2p} + \frac{1}{p} e^{-p} + \left(\frac{1}{p^2} e^{-pt} \Big|_1^2 \right) \\ &= -\frac{2}{p} e^{-2p} + \frac{1}{p} e^{-p} + \frac{1}{p^2} e^{-2p} - \frac{1}{p^2} e^{-p} \end{aligned}$$

$$\begin{aligned} y &= \frac{1}{(p-1)^2} + e^{-2p} \left(\frac{1}{p^2(p-1)^2} - \frac{2}{p(p-1)^2} \right) + e^{-p} \left(\frac{1}{p(p-1)^2} - \frac{1}{p^2(p-1)^2} \right) \\ &= \frac{1}{(p-1)^2} + e^{-2p} \left(\frac{-1}{(p-1)^2} + \frac{1}{p^2} \right) + e^{-p} \left(\frac{1}{p-1} - \frac{1}{p^2} - \frac{1}{p} \right) \end{aligned}$$

$\downarrow$ PFD + simplify

$$y(t) = t e^t + u(t-2) \left((t-2) - (t-2) e^{t-2} \right) + u(t-1) \left(e^{t-1} - (t-1) + 1 \right)$$

* Hopefully no errors *

$$7) \begin{cases} y'' - 5y' + 6y = g(t) \\ y(0) = 0 \quad y'(0) = 0 \end{cases}$$

Let $y = \mathcal{L}[y(t)]$ and $G(p) = \mathcal{L}[g(t)]$. Apply $\mathcal{L}$

$$(p^2 - 5p + 6)y = G$$

$$y = G \left(\frac{1}{(p-3)(p-2)} \right) = G \left(\frac{1}{p-3} - \frac{1}{p-2} \right)$$

$$y(t) = g(t) * (e^{3t} - e^{2t})$$

$$9) f(t) + 2 \int_0^t \cos(t-x) f(x) dx = e^{-t}$$

Apply L.T.

$$F + 2 \left(\frac{p}{p^2 + 1} \right) F = \frac{1}{p-1}$$

$$(1 + \frac{2p}{p^2+1}) F = \frac{1}{p-1}$$

$$\left(\frac{p^2+2p+1}{p^2+1} \right) F = \frac{1}{p-1}$$

$$F = \frac{1}{p-1} \left(\frac{p^2+1}{(p+1)^2} \right) = \frac{1}{2} \left(\frac{1}{p-1} + \frac{1}{p+1} - \frac{2}{(p+1)^2} \right)$$

Apply $\mathcal{L}^{-1}$

$$f(t) = \frac{1}{2} [e^t + e^{-t} - 2te^{-t}]$$

$$8) \begin{cases} y'' + 9y = \delta(x-3\pi) + \cos(3x) \\ y(0) = 0 \quad y'(0) = 0 \end{cases}$$

Let $y(p) = \mathcal{L}[y(x)]$. Apply L.T.

$$(p^2 + 9)y = e^{-3\pi p} + \frac{p}{p^2+9}$$

$$y = e^{-3\pi p} \left(\frac{1}{3} \right) \left(\frac{3}{p^2+9} \right) + \frac{p}{(p^2+9)^2}$$

$$y(x) = \frac{1}{3} u(x-3\pi) \cos(3(x-3\pi)) + x \sin(3x)$$